How the Biopsychosocial Model Frames Chronic Pain Research

A systematic review with mixed-method synthesis of review literature

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Abstract

Background. The biopsychosocial (BPS) model underpins much of chronic pain review literature, yet whether it is genuinely operationalized or merely invoked as a rhetorical label has not been systematically examined.

Methods A pre-registered staged systematic review (OSF DOI: 10.17605/OSF.IO/T4FAM) searched MEDLINE (PubMed) and Web of Science (January 1977–March 2026), applied reproducible deduplication and Stage 1 eligibility rules, extracted Stage 2 abstract-level variables with structured LLM-first coding under schema validation plus deterministic fallbacks, prepared Stage 3 full-text candidates with retrieval-status auditing and manual adjudication checklists, and performed ontology-aligned semantic loading (42 subdomains; transformer embeddings; benchmark-relative domain and co-loading analyses).

Results. Of 6,268 identified records (4,105 after deduplication), 111 met eligibility criteria. Narrative and expert reviews dominated; musculoskeletal pain was most prevalent (44.0%). BPS language appeared at abstract level only in 78.4% of records. Provisional typological coding classified 30.6% as pseudo-BPS or partial signal and 29.7% as potential integrative signal. Semantic loading identified psychological orientation as dominant in 46.8% of records versus 43.2% biological and only 9.9% social; mean social loading (0.318) consistently lagged biological (0.341) and psychological (0.341). Depression, stress, and anxiety dominated the psychological construct profile.

Conclusion. BPS language is ubiquitous but operationally uneven: most reviews demonstrate domain listing rather than mechanistic integration, social factors are systematically underspecified, and the psychological construct profile is dominated by affect labels. These findings have direct implications for reporting standards in BPS-invoking pain reviews.

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1 Introduction

Chronic pain is among the most prevalent and disabling health conditions worldwide, affecting an estimated 20% of adults in Europe and generating substantial personal, societal, and economic burden [Breivik et al., 2006, Rice et al., 2016, Vos et al., 2016]. Over the past five decades its management has been transformed by a fundamental shift: from reductive biomedical causalism toward the biopsychosocial (BPS) model. Introduced by Engel in 1977 as a direct challenge to the Cartesian biomedical tradition, the BPS model proposed that biological, psychological, and social factors interact dynamically to shape the experience and trajectory of illness [Engel, 1977, 1980]. In no clinical domain has this shift been more consequential than in chronic pain research, where the model now underpins clinical guidelines, rehabilitation frameworks, and research agendas across musculoskeletal, neuropathic, and primary pain conditions [Gatchel et al., 2007, Kamper et al., 2015, Nicholas et al., 2019].

The intellectual appeal of the BPS model lies in its scope: it licenses interdisciplinary enquiry, justifies multimodal intervention, and resists the reductionism that proved inadequate for explaining pain chronification. Major theoretical contributions—from gate control theory [Melzack and Wall, 1965] and the neuromatrix [Melzack, 1999] to the fear-avoidance model [Vlaeyen and Linton, 2000] and central sensitization frameworks [Woolf, 2011]—situate naturally within a BPS architecture. Yet the same breadth that makes the model attractive creates a central methodological problem: "biopsychosocial" is semantically broad enough to cover radically different forms of reasoning, from mechanistic triadic integration to mere multi-domain listing, and from theoretically-grounded framing to post-hoc rhetorical labelling.

Conceptual critics have long argued that the BPS model functions as a paradigm whose practical meaning is difficult to specify and verify [Borrell-Carrió et al., 2004, Wade and Halligan, 2017, Crombez, 2015]. When reviews claim BPS framing without defining what integration means or how domain interactions are specified, the model risks becoming a legitimating device rather than an explanatory framework. This concern has been articulated for musculoskeletal pain [Hartvigsen et al., 2018, Foster et al., 2018], chronic primary pain [Treede et al., 2019], and multidisciplinary rehabilitation [Flor et al., 1992], but has never been examined systematically across the entire body of BPS-invoking review literature.

Existing work on how BPS framing operates is fragmentary. Reviews of psychological treatments [Morley et al., 1999, Williams et al., 2020] and multidisciplinary approaches [Kamper et al., 2015] produce consistent outcome evidence but do not interrogate the conceptual architecture of BPS claims. Work on psychological risk factors [Linton, 2000, Pincus et al., 2002] and catastrophizing [Sullivan et al., 1995, Keefe et al., 2004] demonstrates explanatory power of specific constructs but treats the BPS model as background framing. A systematic examination of *how* the model is actually used—where it appears, what function it serves, which domains it covers, and whether integration is specified—has not been conducted.

This review addresses that gap. Drawing on a pre-registered protocol [Van Severen et al., 2026] and a staged, protocol-first design, we examined 111 eligible review articles to determine: (1) how the BPS model is operationalized across chronic pain review literature; (2) the scope, balance, and integration of biological, psychological, and social factors in musculoskeletal reviews; and (3) which psychological constructs and theoretical frameworks are most frequently

invoked. A secondary question addressed conceptual clarity problems in BPS usage.

The review makes three specific contributions: a provisional typological classification of BPS operationalization quality; a structured semantic coding layer that turns abstract-level judgments into auditable machine-readable records; and ontology-aligned semantic loading using embedding-derived domain profiles (42 subdomains). Given that Stage 3 full-text coding remains in progress, findings are reported as a high-fidelity interim synthesis with explicit inference boundaries.

2 Research Questions

Primary Questions

RQ1. How is the biopsychosocial model operationalized in chronic pain review literature.

RQ2. What is the scope, balance, and integration of biological, psychological, and social factors in musculoskeletal pain reviews.

RQ3. Which psychological concepts and theoretical frameworks are invoked in chronic pain review literature.

Secondary Question

SRQ1. What conceptual problems are identified in reviews invoking the biopsychosocial model.

Study Hypotheses

Five a priori hypotheses structured the evaluation: (H1) many reviews will use BPS as a rhetorical or organizing device rather than an integrative explanatory framework; (H2) true mechanistic integration will be rare and domain presentation will be parallel rather than interactive; (H3) psychological factors will dominate while biological and social domains will be minimal or absent in some reviews; (H4) integration statements will typically lack mechanistic pathway detail; (H5) conceptual clarity problems (vague definitions, overlapping constructs) will be common. Evaluation of these hypotheses structures the Results and Discussion.

3 Methods

3.1 Protocol and Reporting

The study was designed as a pre-registered, staged mixed-method systematic review and reported using PRISMA-oriented flow logic [Page et al., 2021, Moher et al., 2009] adapted for conceptual synthesis. The OSF registration defines the governing framework for eligibility, coding, reliability checks, and reconciliation; the implemented pipeline preserves those constraints by separating automated preprocessing from human decision authority at eligibility and adjudication steps.

3.2 Computational Reproducibility

All operational steps were executed through version-controlled command-line stages in the `bps_review` package (Python 3.11 environment), with configuration loaded from the project `config/` files. The run order was database retrieval, deduplication, screening preparation, Stage 1 screening, Stage 2 extraction, Stage 3 preparation, semantic loading, and asset build. Each stage writes structured outputs to fixed `review_stages/` and `paper/assets/tables` locations, enabling exact reruns and file-level audit trails. Deterministic settings were enforced for stochastic procedures (for example, fixed random seed in projection steps).

3.3 Eligibility Criteria

Eligibility criteria were operationalized from the PICOS framework:

- *Population.* Adults ($\geq$ 18 years) with chronic or persistent pain (duration $>$ 3 months), including mixed adult samples where chronic pain was the primary focus.
- *Concept exposure.* Explicit biopsychosocial terminology ("biopsychosocial", "bio-psychosocial", or "bio psycho social") in title or abstract.
- *Outcome.* Operationalization of the BPS model: domain coverage, integration signals, conceptual function, and psychological construct architecture.
- *Study design.* Review articles in any form: systematic review, meta-analysis, scoping review, narrative review, umbrella review, or expert synthesis.

Records were excluded if they were primary studies, protocols without results, editorials, commentaries, letters, conference abstracts, books or book chapters, pediatric-only populations, acute-pain-only studies, animal studies, non-English language sources, or grey literature.

3.4 Information Sources and Search Execution

Searches were executed in MEDLINE (via PubMed API) and Web of Science Core Collection (Starter API). The protocol also specifies PsycINFO (EBSCOhost/EDS); PsycINFO retrieval was unavailable in this run because credentials were not available at execution time and was logged as a protocol deviation. The operational date window was January 1977 to March 2026. Registered search expressions combine biopsychosocial lexical variants with chronic-pain and review-design filters; all query keys, retrieval metadata, and normalized exports are stored in the audit outputs.

3.5 Screening, Deduplication, and Decision Governance

All retrieval outputs were normalized to a common schema before deduplication. Deduplication used title/abstract similarity and identifier checks, then produced a single merged screening set. Stage 1 applied explicit title/abstract decision rules derived from the registered criteria and generated exclusion labels, borderline flags, and reliability materials. In line with the OSF framework, reliability templates target 20% double-screening subsets (capped at 50 for Stage 1/2 and 20 for Stage 3 full-text checks). Discrepancies are resolved by consensus and principal

investigator adjudication. Titles, abstracts, authors, journal, and year remained visible (no masking).

3.6 Stage 2 Extraction and Coding

Stage 2 extracted bibliographic metadata and BPS-operationalization variables from title and abstract for all Stage 1-included records. The implemented extraction stack combined deterministic metadata parsing with structured LLM-first coding using schema-constrained outputs, archived response payloads, and post-hoc validation. Variables included review design, objective category, ICD-11 pain category [Treede et al., 2019, Nicholas et al., 2019], BPS mention location/function, substantive domain indicators, quality-reporting indicators, psychological concepts, framework mentions, conceptual-problem flags, provisional typology, and auditable rationale fields. To avoid lexical inflation, coding rules explicitly disallowed counting the token "biopsychosocial" as sufficient evidence of triadic substantive coverage.

3.7 Stage 3 Full-Text Preparation, Retrieval, and Manual Governance

Stage 3 preparation targeted musculoskeletal and musculoskeletal-unclear records for full-text deep coding (RQ2/RQ3). The pipeline generated a candidate manifest with retrieval source, retrieval status, cached-text path, content-length checks, manual retrieval flags, and manual relevance flags. Open-access retrieval from PubMed Central was attempted first; unresolved records were routed to a manual retrieval queue. A run-level manual relevance checklist and OSF-alignment checklist were exported so that final inclusion relevance and interpretive coding remain human-adjudicated, consistent with the registration's governance model.

3.8 Semantic Loading Analysis

To complement structured coding, we applied an ontology-aligned semantic loading procedure over a shared embedding space. For each record, a composite text (title + abstract + objective) was embedded using `text-embedding-3-small`. Ontology prompts for 42 subdomains (10 biological, 20 psychological, 12 social) were embedded independently; cosine similarities were softmax-normalized to obtain record-level domain loadings and weighted subdomain contributions. Pairwise products (bio×psycho, bio×social, psycho×social) and triadic products were computed to quantify co-loading intensity relative to equal-balance benchmarks. For landscape visualization, joint record-plus-ontology embeddings were projected with PCA-preconditioned t-SNE under fixed random seed settings to preserve rerun stability.

3.9 Synthesis Strategy and Inference Boundaries

The synthesis integrates descriptive statistics (frequencies, cross-tabulations, proportions, and benchmark-relative deviations), provisional typological classification, and qualitative interpretation. Stage 2 assigns one of four provisional operationalization categories: potential integrative signal, multifactorial signal, rhetorical label signal, and pseudo-BPS or partial signal. These categories are explicit interim bridges to the Stage 3 full-text typology (True Integrative BPS, Multifactorial, Pseudo-BPS, Rhetorical BPS). Because Stage 3 retrieval and adjudication are

ongoing, integration-quality claims are interpreted as provisional and bounded to currently available full-text evidence.

3.10 Pre-registration

The review was pre-registered prior to data collection (Van Severen et al., 2026). The registration defines screening, extraction, reliability, reconciliation, and governance expectations; run-specific deviations (for example, source-availability constraints) are logged in reproducible audit outputs and supplementary documentation.

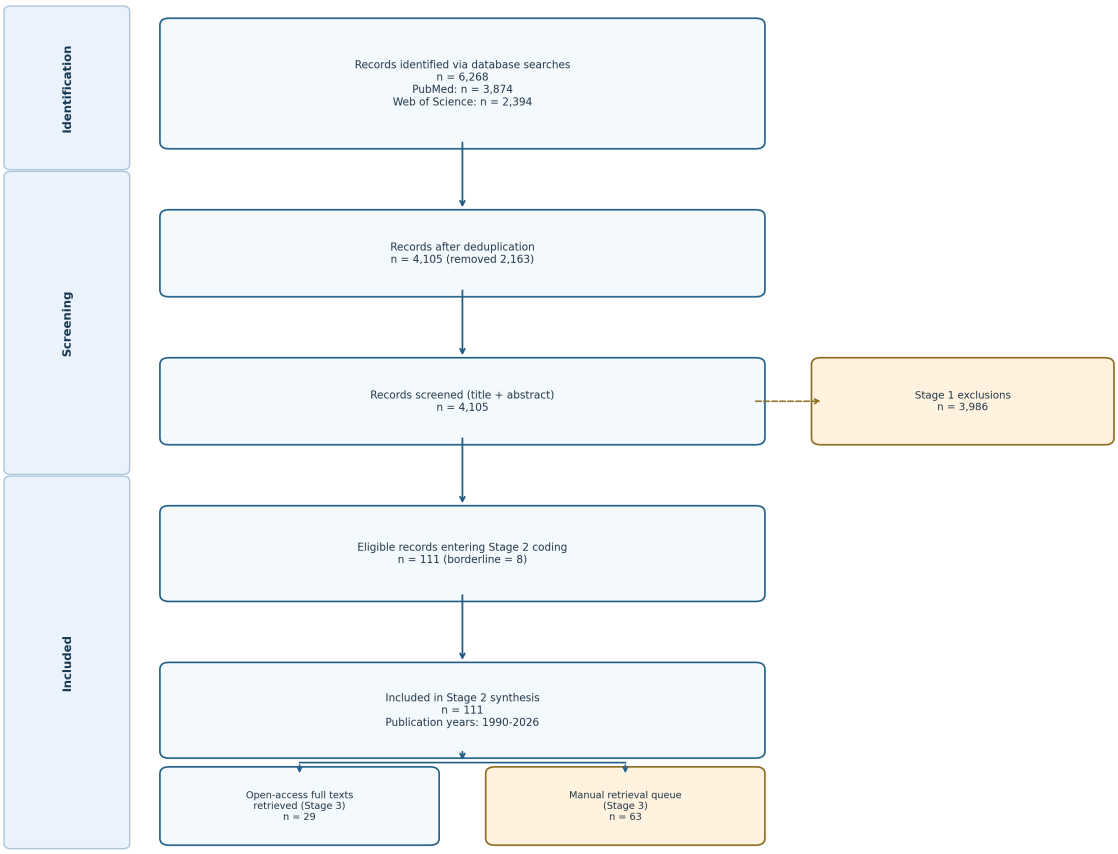
4 Results

4.1 Pipeline Flow and Study Volume

A total of 6268 records were identified across MEDLINE (PubMed) and Web of Science (4105 after deduplication, removing 2163 duplicates). Stage 1 title and abstract screening yielded 111 included, 3986 excluded, and 8 borderline records. Stage 2 abstract-level semantic coding was completed for 111 included reviews spanning publication years 1990–2026. A Stage 3 candidate pool of 92 records was identified for full-text deep coding, of which 29 were retrieved from open-access repositories and 63 remain in the manual retrieval queue. Ontology-aligned semantic loading was applied to all 111 Stage 2 records using openrouter_embedding (openai/text-embedding-3-small).

Table 1. Characteristics of all 111 included reviews.

Figure 1. PRISMA-style study selection workflow from identification through full-text planning.



Note. Three stages: Identification (database searches and deduplication), Screening (title/abstract screening with exclusions), and Included (Stage 2 eligible records and Stage 3 full-text planning). The workflow reflects the operational run covering January 1977–March 2026.

Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Hampf (1990)	This narrative was focused on a biopsychosocial approach to TMJ pain–or looking for keys in the dark.	Narrative	Musculoskeletal	Conceptual	Framework	B, P	Pseudo-BPS
Lurie & Borenstein (1990)	This narrative was focused on PMS is probably a group of entities which include various symptoms that occur during the 7 to 10 days before.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B, P, S	Integrative
Waddell (1992)	This narrative was focused on biopsychosocial analysis of low back pain.	Narrative	Musculoskeletal	Unclear	Unclear	—	Pseudo-BPS
Fordyce (1994)	This narrative was focused on pain and suffering: what is the unit?.	Narrative	Musculoskeletal	Clinical	Framework	P	Pseudo-BPS
Talo et al. (1996)	This narrative was focused on the biopsychosocial disease consequence model in rehabilitation: model development in the Finnish 'work hardening'.	Narrative	Musculoskeletal	Methods	Framework	S	Pseudo-BPS
Ryder (1996)	This narrative was focused on chronic pelvic pain in women may involve more than the gynecologic organ systems.	Narrative	Visceral	Clinical	Framework	—	Pseudo-BPS
Moran et al. (1998)	This narrative was focused on psychiatric problems in chronic facial pain.	Narrative	Headache/ Orofacial	Clinical	Rhetorical	—	Rhetorical
Molin (1999)	This narrative was focused on from bite to mind: TMD–a personal and literature review.	Narrative	Headache/ Orofacial	Conceptual	Framework	P, S	Pseudo-BPS
Wells-Federman (1999)	This narrative was focused on care of the patient with chronic pain: Part I.	Narrative	Mixed/ Unspecified	Clinical	Organizing	B, P, S	Integrative
Loeser (2000)	This narrative was focused on pain and suffering.	Narrative	Mixed/ Unspecified	Conceptual	Rhetorical	B	Rhetorical
Karjalainen et al. (2000)	This systematic was focused on biopsychosocial rehabilitation for upper limb repetitive strain injuries in working age adults.	Systematic	Musculoskeletal	Clinical	Intervention	P, S	Pseudo-BPS
Ferrari (2000)	This narrative was focused on the biopsychosocial model–a tool for rheumatologists.	Narrative	Mixed/ Unspecified	Conceptual	Framework	S	Pseudo-BPS
Wells-Federman (2000)	This narrative was focused on care of the patient with chronic pain: part II.	Narrative	Mixed/ Unspecified	Clinical	Intervention	B, P, S	Multifactorial

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Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Truchon (2001)	This narrative was focused on determinants of chronic disability related to low back pain: towards an integrative biopsychosocial model.	Narrative	Musculoskeletal	Conceptual	Framework	B, P	Pseudo-BPS
Ferrari (2002)	This narrative was focused on prevention of chronic pain after whiplash.	Narrative	Postsurgical	Clinical	Intervention	S	Pseudo-BPS
Goucke (2003)	This narrative was focused on the management of persistent pain.	Narrative	Mixed/ Unspecified	Clinical	Intervention	B, P, S	Multifactorial
Victor & Richeimer (2003)	This narrative was focused on psychosocial therapies for neck pain.	Narrative	Musculoskeletal	Clinical	Framework	P, S	Pseudo-BPS
Waddell (2006)	This narrative was focused on preventing incapacity in people with musculoskeletal disorders.	Narrative	Musculoskeletal	Conceptual	Framework	P, S	Pseudo-BPS
Low et al. (2006)	This narrative was focused on back injuries–getting injured workers back to work.	Narrative	Musculoskeletal	Clinical	Intervention	S	Pseudo-BPS
Bergman (2007)	This narrative was focused on management of musculoskeletal pain.	Narrative	Musculoskeletal	Clinical	Framework	B, P	Pseudo-BPS
Nicholson et al. (2007)	This narrative was focused on psychological risk factors in headache.	Narrative	Headache/ Orofacial	Conceptual	Framework	B, P	Pseudo-BPS
Jensen et al. (2007)	This meta-anal.	Meta-anal.	Musculoskeletal	Mixed	Intervention	B	Pseudo-BPS
Gatchel et al. (2007)	This narrative was focused on the biopsychosocial approach to chronic pain: scientific advances and future directions.	Narrative	Mixed/ Unspecified	Conceptual	Framework	P, S	Pseudo-BPS
Scascighini & Sprott (2008)	This narrative was focused on chronic nonmalignant pain: a challenge for patients and clinicians.	Narrative	Mixed/ Unspecified	Clinical	Intervention	P, S	Pseudo-BPS
Karjalainen et al. (2009)	This systematic was focused on WITHDRAWN: Biopsychosocial rehabilitation for upper limb repetitive strain injuries in working age adults.	Systematic	Musculoskeletal	Clinical	Intervention	P, S	Pseudo-BPS
Buse & Andrasik (2009)	This systematic was focused on behavioral medicine for migraine.	Systematic	Headache/ Orofacial	Clinical	Intervention	B, P, S	Multifactorial

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Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Bay & Liberzon (2009)	This narrative was focused on early stress response: a vulnerability framework for functional impairment following mild traumatic brain injury.	Narrative	Mixed/ Unspecified	Conceptual	Framework	P	Pseudo-BPS
Lee & Tracey (2010)	This narrative was focused on unravelling the mystery of pain, suffering, and relief with brain imaging.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B	Pseudo-BPS
Baeza-Velasco et al. (2011)	This narrative was focused on joint hypermobility syndrome: problems that require psychological intervention.	Narrative	Musculoskeletal	Clinical	Intervention	P, S	Pseudo-BPS
Seshia (2012)	This narrative was focused on chronic daily headache in children and adolescents.	Narrative	Headache/ Orofacial	Clinical	Intervention	B, P, S	Multifactorial
Williams & Cella (2012)	This narrative was focused on medically unexplained symptoms and pain: misunderstanding and myth.	Narrative	Mixed/ Unspecified	Conceptual	Framework	P, S	Pseudo-BPS
Simmons (2012)	This meta-anal.	Meta-anal.	Headache/ Orofacial	Conceptual	Rhetorical	P, S	Rhetorical
Renton et al. (2012)	This narrative was focused on the classification and differential diagnosis of orofacial pain.	Narrative	Headache/ Orofacial	Mixed	Justification	B, S	Pseudo-BPS
Puentedura & Louw (2012)	This systematic was focused on a neuroscience approach to managing athletes with low back pain.	Systematic	Musculoskeletal	Clinical	Intervention	B	Pseudo-BPS
Wilkie & Pransky (2012)	This narrative was focused on improving work participation for adults with musculoskeletal conditions.	Narrative	Musculoskeletal	Clinical	Intervention	B, S	Pseudo-BPS
Woo et al. (2013)	This narrative was focused on evidence-based approach to manage persistent wound-related pain.	Narrative	Mixed/ Unspecified	Clinical	Intervention	B, P	Pseudo-BPS
Pergolizzi et al. (2013)	This narrative was focused on the development of chronic pain: physiological CHANGE necessitates a multidisciplinary approach to treatment.	Narrative	Mixed/ Unspecified	Clinical	Intervention	B, P	Pseudo-BPS
Williams (2013)	This narrative was focused on the importance of psychological assessment in chronic pain.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B, P	Pseudo-BPS

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Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Durham (2013)	This narrative was focused on oral surgery: part 3.	Narrative	Headache/ Orofacial	Clinical	Rhetorical	—	Rhetorical
Shephard et al. (2014)	This narrative was focused on orofacial pain: a guide for the headache physician.	Narrative	Headache/ Orofacial	Clinical	Intervention	P, S	Pseudo-BPS
Phyomaung et al. (2014)	This systematic was focused on are depression, anxiety and poor mental health risk factors for knee pain?.	Systematic	Musculoskeletal	Epidemiol.	Intervention	P	Pseudo-BPS
Matteliano et al. (2014)	This systematic was focused on adherence monitoring with chronic opioid therapy for persistent pain: a biopsychosocial-spiritual approach to.	Systematic	Mixed/ Unspecified	Clinical	Organizing	B, P	Pseudo-BPS
Novy & Aigner (2014)	This narrative was focused on the biopsychosocial model in cancer pain.	Narrative	Cancer-related	Conceptual	Framework	P, S	Pseudo-BPS
Chogle et al. (2014)	This narrative was focused on pediatric IBS: an overview on pathophysiology, diagnosis and treatment.	Narrative	Visceral	Conceptual	Framework	B, P, S	Integrative
Gerdle et al. (2014)	This narrative was focused on chronic musculoskeletal pain: review of mechanisms and biochemical biomarkers as assessed by the microdialysis technique.	Narrative	Musculoskeletal	Conceptual	Framework	B, P, S	Integrative
Hague & Shenker (2014)	This narrative was focused on how to investigate: Chronic pain.	Narrative	Mixed/ Unspecified	Clinical	Rhetorical	P, S	Rhetorical
Pelletier et al. (2015)	This systematic was focused on addressing Neuroplastic Changes in Distributed Areas of the Nervous System Associated With Chronic Musculoskeletal.	Systematic	Musculoskeletal	Clinical	Framework	B, P, S	Integrative
Kent et al. (2015)	This narrative was focused on goal-Directed Resilience in Training (GRIT): A Biopsychosocial Model of Self-Regulation, Executive Functions, and.	Narrative	Mixed/ Unspecified	Clinical	Intervention	B, P	Pseudo-BPS
Obermann et al. (2015)	This narrative was focused on an update on the management of post-traumatic headache.	Narrative	Postsurgical	Clinical	Intervention	P, S	Pseudo-BPS
Thornton & Drummond (2016)	This narrative was focused on current concepts in vulvodynia with a focus on pathogenesis and pain mechanisms.	Narrative	Primary	Clinical	Intervention	B, P, S	Multifactorial

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Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Walk & Poliak-Tunis (2016)	This narrative was focused on chronic Pain Management: An Overview of Taxonomy, Conditions Commonly Encountered, and Assessment.	Narrative	Mixed/ Unspecified	Clinical	Organizing	P	Pseudo-BPS
Cheatle (2016)	This narrative was focused on biopsychosocial Approach to Assessing and Managing Patients with Chronic Pain.	Narrative	Mixed/ Unspecified	Clinical	Intervention	P	Pseudo-BPS
Giummarra et al. (2016)	This systematic was focused on chronic Pain Following Motor Vehicle Collision: A Systematic Review of Outcomes Associated With Seeking or Receiving.	Systematic	Postsurgical	Epidemiol.	Framework	P, S	Pseudo-BPS
Ditre et al. (2016)	This meta-anal.	Meta-anal.	Mixed/ Unspecified	Epidemiol.	Rhetorical	—	Rhetorical
Borsook et al. (2016)	This systematic was focused on reward deficiency and anti-reward in pain chronification.	Systematic	Mixed/ Unspecified	Conceptual	Framework	P	Pseudo-BPS
Zakrzewska (2016)	This systematic was focused on chronic/Persistent Idiopathic Facial Pain.	Systematic	Headache/ Orofacial	Clinical	Intervention	P	Pseudo-BPS
Hooten (2016)	This narrative was focused on chronic Pain and Mental Health Disorders: Shared Neural Mechanisms, Epidemiology, and Treatment.	Narrative	Mixed/ Unspecified	Clinical	Framework	B, P	Pseudo-BPS
Wijma et al. (2016)	This narrative was focused on clinical biopsychosocial physiotherapy assessment of patients with chronic pain: The first step in pain neuroscience.	Narrative	Mixed/ Unspecified	Clinical	Organizing	B, P, S	Integrative
Robinson & Fuchs (2016)	This narrative was focused on allogeneic stem cell transplantation for sickle cell disease.	Narrative	Mixed/ Unspecified	Clinical	Rhetorical	—	Rhetorical
Fillingim (2017)	This narrative was focused on individual differences in pain: understanding the mosaic that makes pain personal.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B, P	Pseudo-BPS
Bardin et al. (2017)	This narrative was focused on diagnostic triage for low back pain: a practical approach for primary care.	Narrative	Musculoskeletal	Clinical	Organizing	—	Pseudo-BPS
Booth et al. (2017)	This narrative was focused on exercise for chronic musculoskeletal pain: A biopsychosocial approach.	Narrative	Musculoskeletal	Clinical	Intervention	B, P, S	Multifactorial

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Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Bartley et al. (2017)	This systematic was focused on predictors of Osteoarthritis Pain: the Importance of Resilience.	Systematic	Musculoskeletal	Conceptual	Framework	S	Pseudo-BPS
Draper-Rodi et al. (2018)	This scoping was focused on identification of prognostic factors and assessment methods on the evaluation of non-specific low back pain in a.	Scoping	Musculoskeletal	Clinical	Organizing	—	Pseudo-BPS
Gagliese et al. (2018)	This systematic was focused on pain, aging and dementia: Towards a biopsychosocial model.	Systematic	Mixed/ Unspecified	Conceptual	Framework	B, P	Pseudo-BPS
Chou et al. (2018)	This systematic was focused on the Global Spine Care Initiative: applying evidence-based guidelines on the non-invasive management of back and neck.	Systematic	Musculoskeletal	Clinical	Intervention	B, P, S	Multifactorial
Martucci & Mackey (2018)	This narrative was focused on neuroimaging of Pain: Human Evidence and Clinical Relevance of Central Nervous System Processes and Modulation.	Narrative	Mixed/ Unspecified	Conceptual	Rhetorical	B, S	Rhetorical
Tai et al. (2018)	This systematic was focused on enriched Environment and Effects on Neuropathic Pain: Experimental Findings and Mechanisms.	Systematic	Neuropathic	Clinical	Rhetorical	B, P	Rhetorical
Brandow & DeBaun (2018)	This narrative was focused on key Components of Pain Management for Children and Adults with Sickle Cell Disease.	Narrative	Mixed/ Unspecified	Clinical	Background	B	Rhetorical
Luchting & Azad (2019)	This narrative was focused on pain therapy for the elderly patient: is opioid-free an option?.	Narrative	Mixed/ Unspecified	Clinical	Intervention	B, P	Pseudo-BPS
Mousavi et al. (2019)	This narrative was focused on low back pain: Moving toward mechanism-based management.	Narrative	Musculoskeletal	Clinical	Intervention	—	Pseudo-BPS
Schweinhardt (2019)	This narrative was focused on where has the 'bio' in bio-psycho-social gone?.	Narrative	Postsurgical	Conceptual	Framework	B, P	Pseudo-BPS
Ford et al. (2019)	This narrative was focused on the Evolving Case Supporting Individualised Physiotherapy for Low Back Pain.	Narrative	Musculoskeletal	Clinical	Intervention	—	Pseudo-BPS
Paley & Johnson (2019)	This systematic was focused on acupuncture for the Relief of Chronic Pain: A Synthesis of Systematic Reviews.	Systematic	Mixed/ Unspecified	Clinical	Background	B, P, S	Multifactorial

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Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Şen et al. (2019)	This systematic was focused on group Psychotherapy With Fibromyalgia Patients.	Systematic	Primary	Clinical	Conclusion	P, S	Rhetorical
Holopainen et al. (2020)	This systematic was focused on physiotherapists' perceptions of learning and implementing a biopsychosocial intervention to treat musculoskeletal.	Systematic	Musculoskeletal	Clinical	Intervention	—	Pseudo-BPS
Greene & Manfredini (2020)	This narrative was focused on treating Temporomandibular Disorders in the 21st Century: Can We Finally Eliminate the "Third Pathway"?.	Narrative	Headache/ Orofacial	Clinical	Justification	S	Pseudo-BPS
Nduwimana et al. (2020)	This meta-anal.	Meta-anal.	Musculoskeletal	Clinical	Intervention	—	Pseudo-BPS
Baxter et al. (2020)	This narrative was focused on six things you need to know about low back pain.	Narrative	Musculoskeletal	Clinical	Intervention	—	Pseudo-BPS
D'Amico et al. (2020)	This scoping was focused on mapping Assessments Instruments for Headache Disorders against the ICF Biopsychosocial Model of Health and Disability.	Scoping	Headache/ Orofacial	Methods	Organizing	P, S	Pseudo-BPS
Xu et al. (2021)	This narrative was focused on pain Management in Neurosurgery: Back and Lower Extremity Pain, Trigeminal Neuralgia.	Narrative	Mixed/ Unspecified	Clinical	Intervention	—	Pseudo-BPS
Argyriou et al. (2021)	This systematic was focused on an updated brief overview on post-traumatic headache and a systematic review of the non-pharmacological.	Systematic	Headache/ Orofacial	Clinical	Intervention	B, P, S	Multifactorial
Najem et al. (2021)	This systematic was focused on religious Beliefs and Attitudes in Relation to Pain, Pain-Related Beliefs, Function, and Coping in Chronic.	Systematic	Musculoskeletal	Clinical	Background	P	Rhetorical
Mustafa et al. (2022)	This narrative was focused on toll-Like Receptor 4 in Pain: Bridging Molecules-to-Cells-to-Systems.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B, P	Pseudo-BPS
Green et al. (2022)	This narrative was focused on persistent Pelvic Pain in Patients With Endometriosis.	Narrative	Visceral	Clinical	Organizing	B	Pseudo-BPS
Edgar et al. (2022)	This narrative was focused on biopsychosocial approach to tendinopathy.	Narrative	Musculoskeletal	Clinical	Intervention	B, P, S	Multifactorial

Continued on next page

Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Ferlito et al. (2022)	This systematic was focused on pain Education in the Management of Patients with Chronic Low Back Pain.	Systematic	Musculoskeletal	Clinical	Intervention	P, S	Pseudo-BPS
Jurak et al. (2023)	This network MA was focused on effects of Multidisciplinary Biopsychosocial Rehabilitation on Short-Term Pain and Disability in Chronic Low Back.	Network MA	Musculoskeletal	Clinical	Intervention	—	Pseudo-BPS
Cormack et al. (2023)	This narrative was focused on the biopsychosocial model is lost in translation: from misrepresentation to an enactive modernization.	Narrative	Musculoskeletal	Conceptual	Framework	S	Pseudo-BPS
Overstreet et al. (2023)	This narrative was focused on the Experience of BIPOC Living with Chronic Pain in the USA: Biopsychosocial Factors that Underlie Racial.	Narrative	Mixed/ Unspecified	Epidemiol.	Framework	B, P, S	Integrative
Włodarczyk et al. (2023)	This narrative was focused on management of Pelvic Pain in Patients with Crohn's Disease-Current Overview.	Narrative	Visceral	Clinical	Background	B	Rhetorical
Gerhart et al. (2023)	This narrative was focused on top Ten Tips Palliative Care Clinicians Should Know About Behavioral Pain Management for Persistent Pain.	Narrative	Mixed/ Unspecified	Clinical	Framework	P	Pseudo-BPS
Frazier et al. (2023)	This narrative was focused on diagnosis and Management of Cyclic Vomiting Syndrome: A Critical Review.	Narrative	Headache/ Orofacial	Clinical	Intervention	B, P, S	Multifactorial
Carey & McClurg (2023)	This systematic was focused on evaluation and Medical Management of Chronic Pelvic Pain.	Systematic	Visceral	Clinical	Intervention	—	Pseudo-BPS
Rahman et al. (2023)	This narrative was focused on clinical Diagnosis and Treatment of Chronic Pain.	Narrative	Mixed/ Unspecified	Clinical	Organizing	—	Pseudo-BPS
Ruimonte-Crespo et al. (2023)	This narrative was focused on aerobic Exercise and Neuropathic Pain: Insights from Animal Models and Implications for Human Therapy.	Narrative	Neuropathic	Clinical	Rhetorical	B	Rhetorical

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Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Shil et al. (2024)	This narrative was focused on a clinical approach to the investigation and management of long COVID associated neuropathic pain.	Narrative	Neuropathic	Clinical	Intervention	B	Pseudo-BPS
Warzocha et al. (2024)	This systematic was focused on etiologic Factors of Temporomandibular Disorders: A Systematic Review of Literature Containing Diagnostic Criteria.	Systematic	Headache/ Orofacial	Epidemiol.	Rhetorical	P, S	Rhetorical
Polonowita et al. (2024)	This narrative was focused on construction of the chronic temporomandibular disorder patients: the association between neural and psychological.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B, P, S	Integrative
Lawan et al. (2024)	This meta-anal.	Meta-anal.	Musculoskeletal	Methods	Unclear	S	Pseudo-BPS
Couëpel et al. (2024)	This scoping was focused on what are the functional and clinical characteristics shared by fibromyalgia and low back pain?.	Scoping	Primary	Clinical	Organizing	B, P, S	Integrative
Dehghan et al. (2024)	This narrative was focused on deciphering pain: molecular mechanisms and neurochemical pathways-challenges and future opportunities.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B	Pseudo-BPS
Alonso-Sal et al. (2024)	This systematic was focused on effectiveness of Physical Activity in the Management of Nonspecific Low Back Pain.	Systematic	Musculoskeletal	Clinical	Intervention	B, P	Pseudo-BPS
Rometsch et al. (2025)	This systematic was focused on chronic pain in European adult populations: a systematic review of prevalence and associated clinical features.	Systematic	Mixed/ Unspecified	Epidemiol.	Background	P, S	Rhetorical
Souza et al. (2025)	This scoping was focused on conversational Agents to Support Pain Management.	Scoping	Mixed/ Unspecified	Clinical	Background	P	Rhetorical
Kubala et al. (2025)	This systematic was focused on biopsychosocial and physical factors associated with firefighter musculoskeletal disorders.	Systematic	Musculoskeletal	Epidemiol.	Organizing	P, S	Pseudo-BPS
Potthoff et al. (2025)	This systematic was focused on social dimensions as explanatory approaches for the development of chronic pain: a meta-ethnography of qualitative.	Systematic	Mixed/ Unspecified	Conceptual	Framework	B, P, S	Integrative

Continued on next page

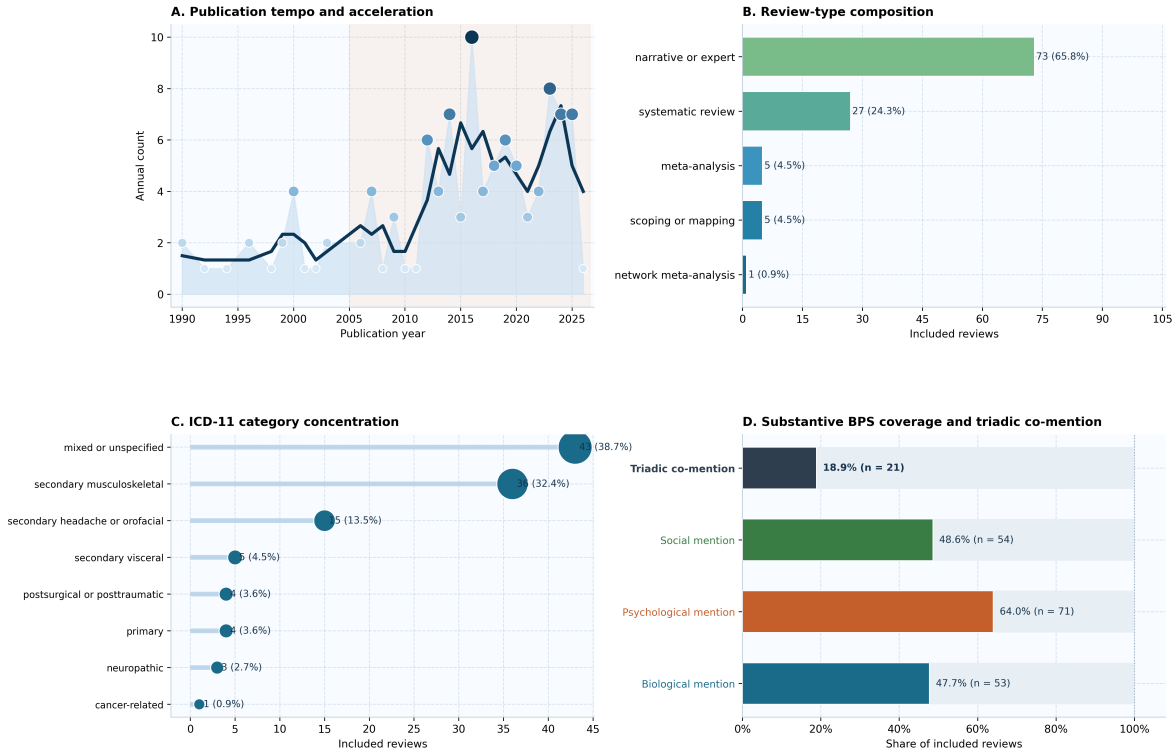
Reference	Description	Type	Pain Condition	Objective	BPS Role	Dom.	Typology
Mueller et al. (2025)	This systematic was focused on advances in Our Understanding and Treatment of Nociplastic Pain.	Systematic	Primary	Clinical	Intervention	B	Pseudo-BPS
Strien & Hollmann (2025)	This narrative was focused on pain Perception and Modulation: Fundamental Neurobiology and Recent Advances.	Narrative	Mixed/ Unspecified	Conceptual	Framework	B, P	Pseudo-BPS
Kohlert et al. (2025)	This scoping was focused on existential therapy for treating chronic pain.	Scoping	Mixed/ Unspecified	Clinical	Intervention	P	Pseudo-BPS
Boon et al. (2026)	This narrative was focused on dual Experience and Management of Pain in Older Adults and Their Caregivers.	Narrative	Mixed/ Unspecified	Clinical	Framework	S	Pseudo-BPS

Note. BPS Role: primary declared functional role of BPS language in the review. Dom.: substantive domain mentions in title/objective/abstract after excluding lexical BPS token matches (B = biological, P = psychological, S = social). Typology: provisional BPS operationalization class from Stage 2 abstract-level coding. Description: concise past-tense one-sentence summary generated from cached full-text objective statements when available; otherwise from objective text, abstract, and finally title fallback. Full DOI/PMID links are provided in the generated included-review catalog.

4.2 Temporal and Structural Characteristics of Included Reviews

The included corpus spanned publication years 1990 to 2026, with a marked acceleration in BPS-invoking review output after 2005. Narrative and expert reviews were the dominant review type (73 of 111; 65.8%), consistent with the predominantly conceptual and clinical orientation of BPS-invoking pain literature; systematic reviews comprised 27 records (24.3%), with scoping reviews, meta-analyses, and one network meta-analysis accounting for the remainder. By ICD-11 classification, mixed or unspecified chronic pain was the most prevalent category (43 records; 38.7%), followed by musculoskeletal pain (36; 32.4%), headache or orofacial pain (15; 13.5%), and visceral pain (5; 4.5%). The remaining 12 records span primary chronic pain, postsurgical, neuropathic, and cancer-related categories. Temporal trends, review-type composition, ICD-11 distribution, and BPS domain coverage are shown in Figure 2.

Figure 2. Descriptive structure of the included evidence base across four dimensions.



Note. Panel A shows annual publication counts ($n = 111$) with a 3-year rolling trend line and post-2005 growth shading. Panel B shows review-type distribution with frequency labels. Panel C shows concentration across ICD-11 categories. Panel D compares biological, psychological, and social substantive-domain coverage plus overall triadic co-mention frequency across all 111 records.

4.3 Biopsychosocial Operationalization

Mention Location and Functional Role

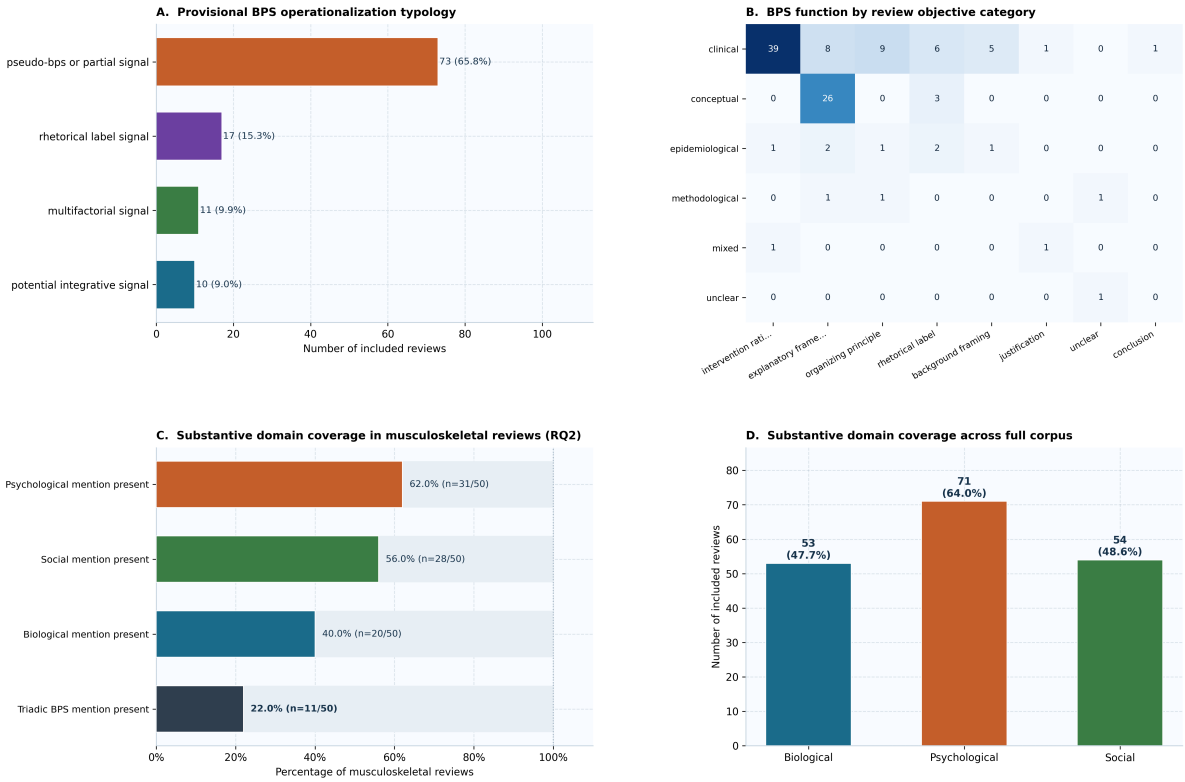
BPS terminology appeared predominantly in the abstract without title mention in 87 records (78.4%), indicating that for most reviews the model functions as methodological or conceptual context rather than as the primary subject of inquiry. A further 18 records (16.2%) mentioned BPS in both title and abstract, and 6 (5.4%) in the title only. The most common declared func-

tional role was intervention rationale (41 records; 36.9%), followed by explanatory framework (37; 33.3%), organizing principle (11; 9.9%), rhetorical label (11; 9.9%), and background framing (6; 5.4%). Many records using the explanatory framework label do not provide mechanistic elaboration of domain interactions, providing partial support for study hypothesis H4.

Provisional Typological Classification

The provisional typological scheme identified pseudo-BPS or partial signal as the dominant category (73 records; 65.8%), followed by rhetorical label signal (17; 15.3%), multifactorial signal (11; 9.9%), and potential integrative signal (10; 9.0%). Thus, approximately four in five records fall below the potential-integrative threshold at abstract level. These typological estimates are provisional pending Stage 3 full-text adjudication and address study hypotheses H1 and H2. Figure 3 (Panel A) visualizes the distribution; Panel B shows the cross-tabulation of BPS function by objective category.

Figure 3. Biopsychosocial operationalization structure: typology, BPS function, musculoskeletal domain coverage, and full-corpus domain mentions.



Note. Panel A: provisional typological distribution (RQ1); categories are assigned by Stage 2 coding and remain provisional pending Stage 3 adjudication. Panel B: cross-tabulation of BPS functional role by objective category; cell values are record counts. Panel C: substantive-domain coverage within the 50 musculoskeletal reviews (RQ2); bars show percentage of musculoskeletal records with each domain/triadic mention. Panel D: overall biological, psychological, and social substantive-domain frequencies across all 111 included records.

BPS Operationalization. Across 111 included reviews, BPS terminology appeared predominantly in the abstract without title mention (dominant location: abstract only; 87 records, 78.4%). The most common declared functional role was *intervention rationale* (36.9%), followed

by explanatory framework (33.3%), together accounting for 70.2% of the corpus. Provisional typological classification: potential integrative signal 9.0%, multifactorial signal 9.9%, pseudo-BPS or partial signal 65.8%, rhetorical label signal 9.9%. This distribution provides empirical support for the pre-specified hypothesis that BPS language is often used as a rhetorical or contextualizing device rather than as a specification of triadic mechanistic integration. Evidence is visualized in Figure 3 (Panels A and B).

Scope, Balance, and Integration in Musculoskeletal Reviews. Using the substantive domain-coding layer (lexical BPS token matches excluded), coverage within the 50 musculoskeletal reviews was: biological 20/50 (40.0%), psychological 31/50 (62.0%), social 28/50 (56.0%). Simultaneous triadic co-mention was present in 11/50 musculoskeletal reviews (22.0%). Across all 111 Stage 2 records, triadic co-mention was 21/111 (18.9%). Ontology-aligned semantic loading (covering 111 records) identified psychological orientation as dominant in 52 records (46.8%), biological in 48 (43.2%), and social in 11 (9.9%). Mean domain loadings were: biological 0.341, psychological 0.341, social 0.318. Taken together, the two layers indicate divergence between domain naming and semantic weight: social content is often present but comparatively light in semantic centrality. Evidence is in Figure 3 (Panel D), Figure 4, and Figure 5.

Psychological Concepts and Frameworks. Stage 2 abstract-level extraction identified the dominant psychological construct cluster as depression, stress, anxiety, which together account for the majority of detected concept occurrences. Theory-specific constructs with established BPS relevance—catastrophizing, fear-avoidance, self-efficacy, illness perception—appear at considerably lower frequencies in abstracts, consistent with the hypothesis that BPS pain reviews emphasize general affect labels over specific cognitive-behavioral mechanisms. Ontology-based subdomain loading shows that within the psychological domain, constructs mapped to cognitive-behavioral and third-wave therapy subdomains carry the highest relative loading weight, while fear-avoidance and self-efficacy subdomains show lower absolute loading—a pattern that may reflect the differential between treatment-oriented and mechanistic research orientations. Full-text Stage 3 coding will map construct definitions, theoretical framework membership, and hierarchical relationships. Evidence is in Figure 4 (Panel A) and the supplementary semantic tables.

4.4 Scope, Balance, and Integration in Musculoskeletal Reviews

Addressing RQ2, the analysis focused on the 50 reviews classified as chronic secondary musculoskeletal pain.

Structured Domain Coverage

Domain mention frequencies for the musculoskeletal subset are illustrated in Figure 3 (Panel C). In the substantive-domain coding layer (which excludes lexical BPS token matches), social factors were present in 28 of 50 reviews (56.0%), psychological factors in 31 of 50 (62.0%), and biological factors in 20 of 50 (40.0%). Triadic mention was present in 11 of 50 musculoskeletal reviews (22.0%). The musculoskeletal subset therefore shows moderate two-domain coverage but limited explicit triadic integration at abstract level, leaving study hypotheses H2 and H3 partially satisfied at this interim stage.

The domain mention pattern across the full corpus of 111 reviews is illustrated in Figure 3 (Panel D). Substantive-domain frequencies were psychological 64.0%, social 48.6%, and biological 47.7%, with triadic co-mention at 18.9%. The main structural deficit at abstract level is therefore not absence of a single domain, but the recurrent failure to combine all three domains in a clearly articulated integrative logic.

Semantic Domain Loading and the Social Deficit

Ontology-aligned semantic loading provided a continuous, text-derived perspective on domain balance. Across all 111 included records, psychological orientation was semantically dominant in 46.8% of records, biological in 43.2%, and social in only 9.9% (supplementary Table 3 and Figure 4). Mean domain loadings were: biological 0.341, psychological 0.341, social 0.318. The social loading deficit—despite substantial social naming in structured coding—constitutes a robust signal that social factors remain comparatively underweighted in semantic centrality, consistent with study hypothesis H3 [Dörner et al., 2011, Harkness et al., 2005, Edwards et al., 2016].

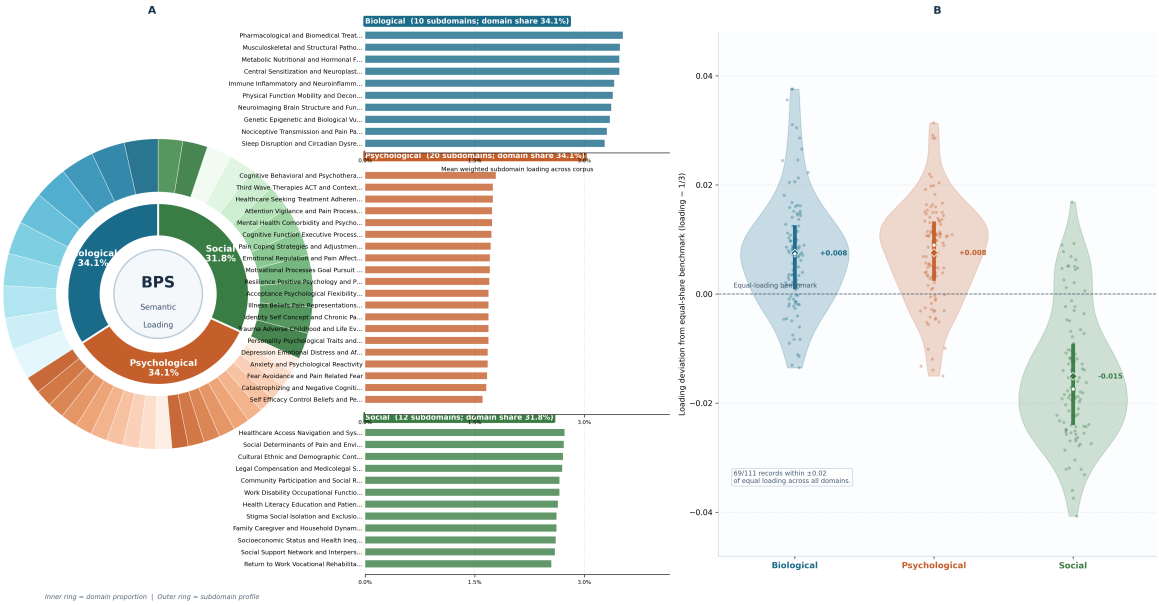
Figure 4 (Panel A) shows the two-layer domain hierarchy: the inner ring presents domain-level loading proportions and the right panel resolves each domain into its 42 ontology subdomains with within-domain normalized loading bars. Within the biological domain, pharmacological/biomedical treatment and central sensitization subdomains carry the highest relative loading. Within psychology, cognitive-behavioral and third-wave therapy subdomains lead, followed by healthcare-seeking and attention/vigilance constructs. Within social, healthcare access and social determinants subdomains rank highest. Panel B reframes the semantic-loading result as benchmark-relative deviations: biological and psychological loadings sit slightly above the equal-share benchmark on average, whereas social loading is shifted below it. Thus the key signal is not large dispersion across BPS space but a repeated, corpus-wide social shortfall.

Pairwise and Triadic Co-loading

Pairwise loading products measure the co-occurrence of two domains within the same semantic space of a record. Figure 5 shows that the bio×psychological term is shifted above the equal-balance benchmark, whereas the bio×social and psychological×social terms lie slightly below it; the triadic product sits only marginally below its equal-share benchmark. The corpus-wide signal is therefore not strong overall imbalance, but a subtle and repeated redistribution of semantic weight away from social content and toward the bio–psychological axis.

The joint embedding landscape (Figure 5, Panel A) situates the included reviews in the joint semantic space of record embeddings, domain anchors, and ontology subdomains. The three ontology anchor fields all lie outside the densest record manifold, but the social anchor is the most spatially distant from that manifold, converging with the loading results: social constructs are present in the ontology and mentioned in the corpus, yet comparatively few reviews organize their semantic centre of mass around them.

Figure 4. Two-layer ontology semantic loading across the BPS hierarchy and benchmark-relative domain deviation distributions.



Note. Panel A: left sub-panel shows the two-ring sunburst (inner ring = domain-level loading proportions; outer ring = 42 subdomain segments without text to avoid overlap); right sub-panel shows mean weighted subdomain loading percentages (absolute, not within-domain normalized) grouped by domain, with domain-share labels. Segment and bar magnitudes are proportional to corpus-level mean weighted loading across 111 records. Panel B: each domain's distribution of deviation from the equal-share benchmark (loading − 1/3), with violin density, raw jittered points, and mean (diamond) and median markers.

4.5 Psychological Concepts and Frameworks

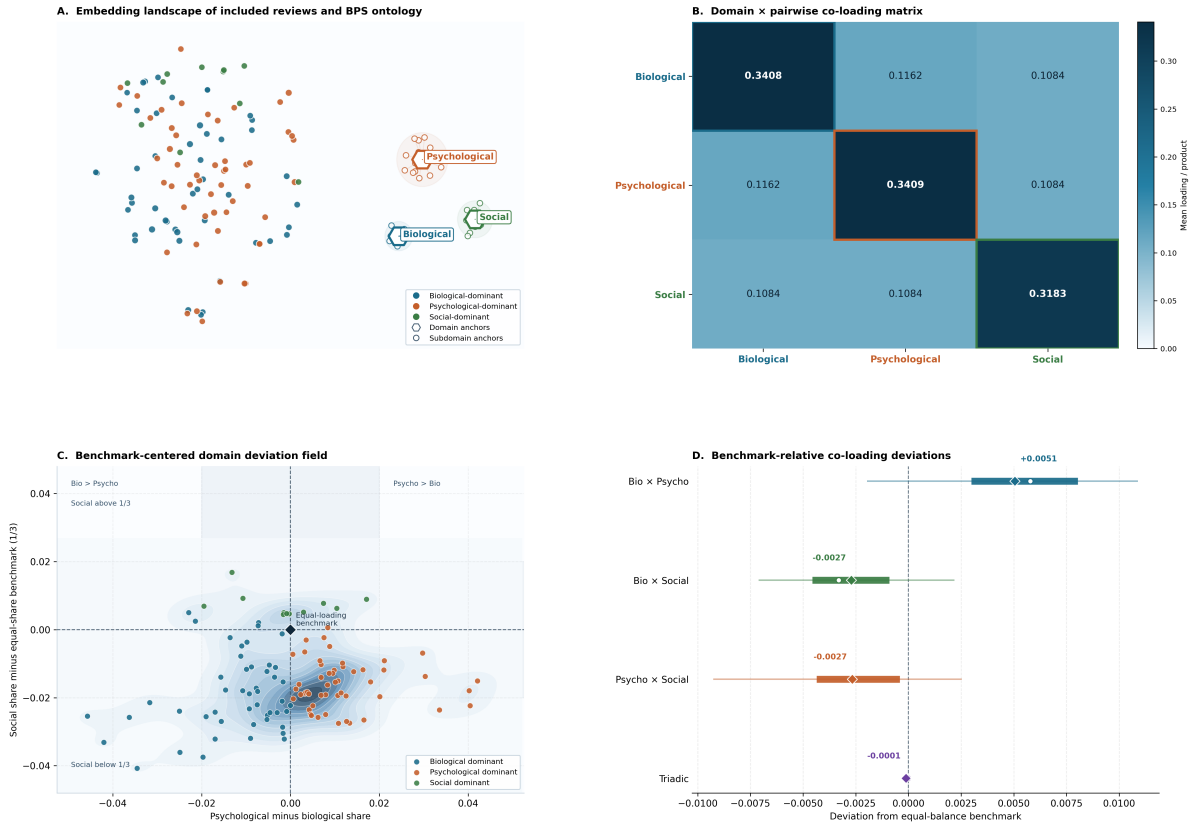
Addressing RQ3, Stage 2 abstract-level extraction identified the most frequently occurring psychological construct terms. The concept profile reveals a markedly concentrated pattern: depression (27.5%), stress (23.8%), and anxiety (20.0%) together account for more than 70% of detected concept occurrences. Coping strategies (8.8%), catastrophizing (5.0%), and fear-avoidance (5.0%) follow as a secondary cluster, while self-efficacy (3.8%), resilience (2.5%), acceptance (2.5%), and illness perception (1.2%) appear at lower frequencies.

This distribution supports study hypothesis H5 about conceptual clarity. The dominance of general affect labels (depression, anxiety, stress) relative to theoretically specific constructs (catastrophizing, fear-avoidance, self-efficacy) suggests that chronic pain review literature characterizes the psychological domain through symptom burden rather than through the BPS-grounded motivational, cognitive, and self-regulatory mechanisms that have the strongest theoretical support [Vlaeyen and Linton, 2000, 2012, McCracken and Vowles, 2014, Bandura, 1977, Sullivan et al., 1995]. Stage 3 full-text coding will map construct definitions, hierarchical relationships, and theoretical framework membership for the musculoskeletal subset.

4.6 Conceptual Problems in BPS Usage

Conceptual Problems in BPS Usage. Structured semantic extraction of conceptual problem flags from the Stage 2 coding layer identified *parallel_listing_without_integration* as the

Figure 5. Joint embedding landscape and ontology-based pairwise and triadic semantic integration loadings.



Note. Panel A: embedding landscape of 111 included reviews (filled circles, coloured by dominant domain), domain anchors (hexagons), and 42 subdomain anchors (open circles connected to parent domains); subdomain anchors are domain-clustered for interpretability while preserving within-domain geometry after projection; background density and contour lines show the review manifold; point size scales with dominant loading strength. Projection uses PCA-preconditioned t-SNE on the joint record-plus-ontology embedding space. Panel B: symmetric domain × pairwise co-loading intensity matrix (cell values = means). Panel C: benchmark-centered deviation field mapping each record onto the psycho−bio axis (horizontal) and social benchmark deviation (vertical), with shaded near-balance bands around zero. Panel D: benchmark-relative co-loading deviations for each pairwise and triadic term (observed minus equal-balance benchmark); bars show IQR, diamonds show means.

most frequent provisional flag. Common patterns include invocation of BPS framing without operational definition of domain interactions, parallel domain listing without integration statements, tokenistic treatment of the social domain, and inconsistent mapping between stated BPS theoretical framework and empirical content. These flags are hypothesis-generating signals for Stage 3 full-text adjudication and should not be treated as confirmed findings.

4.7 Full-Text Retrieval Status and Stage 3 Readiness

Of the 92 records identified as Stage 3 candidates (musculoskeletal and unspecified pain reviews), 29 (31.5%) were retrieved as open-access full texts from PubMed Central, while 63 (68.5%) remain in the manual retrieval queue. The 29 retrieved texts are ready for full-text deep coding once the Stage 3 codebook has been finalized through pilot testing. Completion of the manual retrieval queue is the principal dependency for full inferential closure on RQ2. To align with the

OSF protocol requirement that eligibility and coding remain human-adjudicated, all candidate records are additionally listed in a manual relevance audit sheet with reviewer and adjudication fields.

5 Discussion

5.1 The BPS Model as Label versus Framework

The central finding is an asymmetry between the frequency of BPS language use and the depth of its operationalization. Across 111 included reviews, provisional typological coding classified 65.8% as pseudo-BPS or partial-signal and 15.3% as rhetorical label signal, while only 9.0% reached a potential integrative signal and 9.9% a multifactorial signal. The resulting picture indicates that explicit triadic integration signals are comparatively uncommon at abstract level. The concern is not strategic dishonesty. BPS language is semantically broad enough to invoke without specifying what integration entails: which pathways connect the domains, in which directions, at which time scales. The provisional typology developed here gives this critique empirical grip. A record is coded as potential integrative signal only if it (a) covers all three domains and (b) declares an explanatory framework or organizing principle function—the minimum abstract-level signal of intended integration. That roughly four in five records fail this criterion is a precise, auditable claim, not a subjective impression [Borrell-Carrió et al., 2004, Wade and Halligan, 2017]. This finding provides confirmatory evidence for study hypothesis H1.

5.2 The Social Domain Deficit

Perhaps the most striking finding is the systematic underrepresentation of social factors across both structured abstract-level coding and semantic loading. In the substantive coding layer, social factors were present in 56.0% of musculoskeletal records and 48.6% of the full corpus, yet semantic loading ranked social as dominant in only 9.9% of records (versus 46.8% psychological, 43.2% biological). The mean social loading (0.318) was lower than both biological (0.341) and psychological (0.341). The pattern holds across methods.

This social loading deficit is interpretable as a systematic pattern in which the "S" of BPS is lexically present but conceptually thin—named without the mechanistic or operational specification that biological factors receive through neuroimaging, pharmacology, or genetics, and psychological factors through standardized measurement instruments and theoretical models [Dörner et al., 2011, Harkness et al., 2005, Edwards et al., 2016]. Social determinants (socioeconomic status, occupational context, medicolegal systems, cultural factors) appear in review discourse as labels rather than as substantively operationalized constructs with defined pathways. This confirms study hypothesis H3. Stage 3 integration coding will allow this to be tested against full-text evidence, but the convergence between two independent measurement strategies is a robust preliminary signal.

5.3 Psychological Construct Concentration and the Affect Paradox

The psychological concept distribution—dominated by depression, anxiety, and stress, with limited representation of specific cognitive and self-regulatory constructs—reveals a tension in the pain psychology literature. The theoretical foundations of BPS-aligned pain psychology emphasize specific mechanisms: catastrophizing as a cognitive amplifier of pain [Sullivan et al., 1995]; fear-avoidance as an operant-conditioning pathway from acute to chronic pain [Vlaeyen and Linton, 2000, 2012]; illness beliefs as interpretive frames that shape help-seeking and treatment adherence; and self-efficacy as a cross-cutting mediator of functional outcomes [Bandura, 1977]. Yet the abstract-level concept profile suggests that emotionally oriented general labels dominate how psychological factors are represented in review summaries.

Two interpretations are possible. First, this may be a reporting artefact: abstracts may prioritize outcome-relevant affect labels (depression, anxiety) that are measured by standard instruments, while theory-specific constructs are detailed in the full text. Second, the pattern may reflect a genuine preponderance of affect-focused framing in BPS pain reviews, consistent with meta-analytic findings showing that CBT and interdisciplinary interventions primarily target depressive and anxiety symptoms [Morley et al., 1999, Williams et al., 2020]. Distinguishing these explanations—artefact or signal—is a primary motivation for Stage 3 deep coding.

5.4 Semantic Loading as a Methodological Contribution

The ontology-aligned semantic loading procedure developed here extends structured abstract-level content coding in three ways. First, it is continuous and multi-dimensional: each record receives a domain loading profile rather than a binary domain-mention indicator, capturing graded orientation rather than presence/absence. Second, the subdomain resolution (42 categories) maps specific research constructs within each domain, allowing detection of which biological subdomains (e.g., pharmacological treatment vs. central sensitization) or psychological subdomains (e.g., CBT approaches vs. fear-avoidance) dominate a record's semantic space. Third, the pairwise product terms measure co-loading intensity as an approximation of conceptual overlap between domains, providing a text-derived integration signal that complements the binary triadic co-mention indicator.

The social deficit finding illustrates the procedure's sensitivity: while structured abstract-level coding still shows near-universal social mention, semantic loading shows social dominant in only 9.9% of records, exposing the gap between mentioning and substantively organizing a review around social factors. This methodological contribution—surfacing the qualitative-quantitative divergence between surface-level mention and semantic orientation—is generalizable to other conceptual systematic reviews in pain science and beyond.

5.5 Methodological Limitations

The principal limitation is the interim status of the synthesis. Stage 3 full-text coding remains incomplete, constraining inferential closure on integration quality, construct-level mapping, and balance assessment. The absence of completed manual retrieval for 57 records limits Stage 3 candidate coverage. PsycINFO was not available for this run, which may underrepresent re-

views indexed primarily in psychology databases. The provisional typological scheme relies on abstract-level signals and may misclassify records whose integration quality is only apparent in the full text. Finally, the semantic loading procedure is sensitive to ontology term selection; the current 42-subdomain scheme reflects one operationalization among possible alternatives, and results should be interpreted relative to this specific ontological framing.

This synthesis represents a high-fidelity interim analysis pending Stage 3 full-text coding and dual-coder adjudication. The search covered MEDLINE and Web of Science; PsycINFO records were not available for this run due to credential constraints, which may underrepresent reviews indexed primarily in psychology databases. Stage 3 full-text retrieval is incomplete (53/87 in manual queue), constraining inferential closure on domain integration quality. The provisional typology relies on abstract-level signals and may misclassify records whose integration quality is apparent only in full text. Semantic loading is sensitive to ontology term selection; the 42-subdomain scheme reflects one operationalization among possible alternatives.

5.6 Implications for Research Practice and Reporting Standards

The findings have actionable implications. Future chronic pain reviews that invoke the BPS model should provide explicit reporting of: which domains are covered; how domain interactions are conceptualized; what mechanistic pathways are proposed; and whether theoretical framework claims are supported by the review's empirical content. A practical proposal is that journals adopt mandatory reporting prompts distinguishing rhetorical from substantive BPS integration, modelled on existing PRISMA extensions [Page et al., 2021]. Editorial and peer-review practices that currently accept BPS invocation as self-evidently legitimate would benefit from standardized criteria distinguishing descriptive domain listing from mechanistic triadic integration.

6 Conclusion

This systematic review provides the first comprehensive empirical map of how the biopsychosocial model is operationalized across 111 eligible chronic pain review articles spanning 1990 to 2026. The evidence indicates that BPS language is ubiquitous but operationally uneven: many reviews still demonstrate domain listing or partial coverage rather than mechanistic triadic integration; social factors are consistently underspecified relative to psychological and biological ones in the semantic-loading layer; and the psychological construct profile is dominated by general affect labels rather than the specific cognitive and self-regulatory mechanisms that BPS-aligned theories emphasize. Convergent evidence from two independent methods—structured Stage 2 coding and ontology-aligned semantic loading—strengthens confidence in the social deficit and psychological concentration findings. These results are currently provisional, pending Stage 3 full-text coding of the musculoskeletal subset and dual-coder adjudication. Final inferential claims about integration quality and construct-level framework architecture await completion of manual retrieval and the PsycINFO search. The staged, auditable pipeline and reproducible ontology-based semantic analysis developed here provide the infrastructure for that completion and a template for future conceptual systematic reviews in pain science.

Data and Code Availability

Analysis code, pre-registration materials, and supplementary data are openly available.
Pre-registration (protocol, search strings, codebooks, and audit trail): https://osf.io/t4fam/overview (DOI: 10.17605/OSF.IO/T4FAM; Van Severen et al. 2026).
Code repository (full pipeline, embedding analysis, and figure generation): https://github.com/stvsever/SystematicReview_on_BioPsychoSocialModel_in_ChronicPainReseach.

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A Supplementary Methods

Supplementary Methods Note. Protocol alignment and reliability.
Operational pipeline stages preserve the pre-registered sequencing while logging deviations explicitly. Stage 2 uses structured LLM-based abstract coding with deterministic metadata fields and archived batch outputs; this augments rather than replaces final adjudication. Dual-coder templates are generated for Stage 1, Stage 2, and Stage 3 reliability checks, consistent with the registered subset reliability design. Current reliability file status. stage1. status=not_ready, n=0, percent_agreement=, kappa= stage2. status=not_ready, n=0, percent_agreement=, kappa= stage3. status=not_ready, n=0, percent_agreement=, kappa=
Semantic loading note. Ontology-aligned embeddings generated through OpenRouter.

B Supplementary Semantic Tables

Table 3. Ontology-based semantic loading summary across BPS domains.

domain	mean_cosine	mean_loading	dominance_n
biological	0.5222	0.3408	48
psychological	0.5224	0.3409	52
social	0.4534	0.3183	11

Note. Method=openrouter_embedding; model=openai/text-embedding-3-small. Mean loading uses softmax-normalized cosine similarity.

Table 4. Pairwise and triadic semantic integration summary.

bio_psych_mean	bio_social_mean	psych_social_mean	triadic_product_mean
0.116200	0.108400	0.108400	0.036900

Note. Pairwise scores are products of normalized domain loadings per record.

The full subdomain loading summary (42 subdomains) is archived at:
`paper/assets/tables/semantic_subdomain_summary.csv`.

C Supplementary Catalog of Included Reviews

The complete Stage 2 included review catalog is provided as a machine-readable supplementary file: `paper/assets/tables/included_review_catalog.csv`. Key fields are described in the registered codebook available at the repository listed in the Data and Code Availability section.